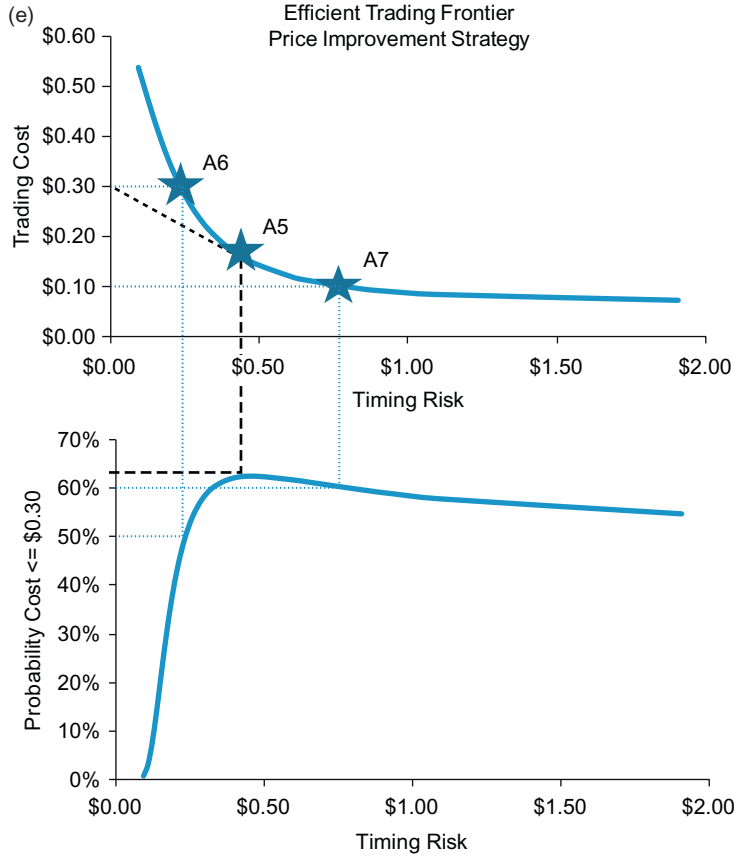




■ Figure 8.2 (e) Price Improvement Derivation.

2. Minimize Cost with Risk Constraint

Our second criterion is to minimize cost for a specified quantity of risk. The risk constraint is often specified by the portfolio manager or by firm mandate that will not allow risk to exceed the specified level $\mathfrak{R}^*$. The optimization is:

$$\begin{aligned} & \text{Min Cost} \\ & \text{s.t. } TR' = \mathfrak{R}^* \end{aligned} \quad (8.15)$$

Example: An investor looking to minimize market impact cost (not including price appreciation) subject to a specified level of timing risk will determine the optimal trading rate through minimizing the following equation:

$$\begin{aligned} & \text{Min: } b_1 \cdot I' \cdot \alpha + (1 - b_1) \cdot I' \\ & \text{s.t. } \sigma \cdot \sqrt{\frac{1}{250} \cdot \frac{1}{3} \cdot \frac{X}{ADV} \cdot \alpha^{-1} \cdot P_0} = \mathfrak{R}^* \end{aligned} \quad (8.16)$$